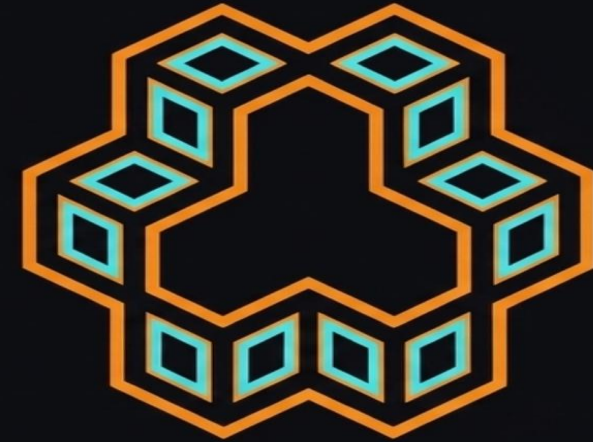


Bank Customer Churn Analysis

Predictive Modeling using Neural Networks :

- Student: Mohammad Hossein Bakhshian
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- Course: Introduction to Artificial Intelligence
- Professor: Dr. Mahdi Aliyari Shooredeli
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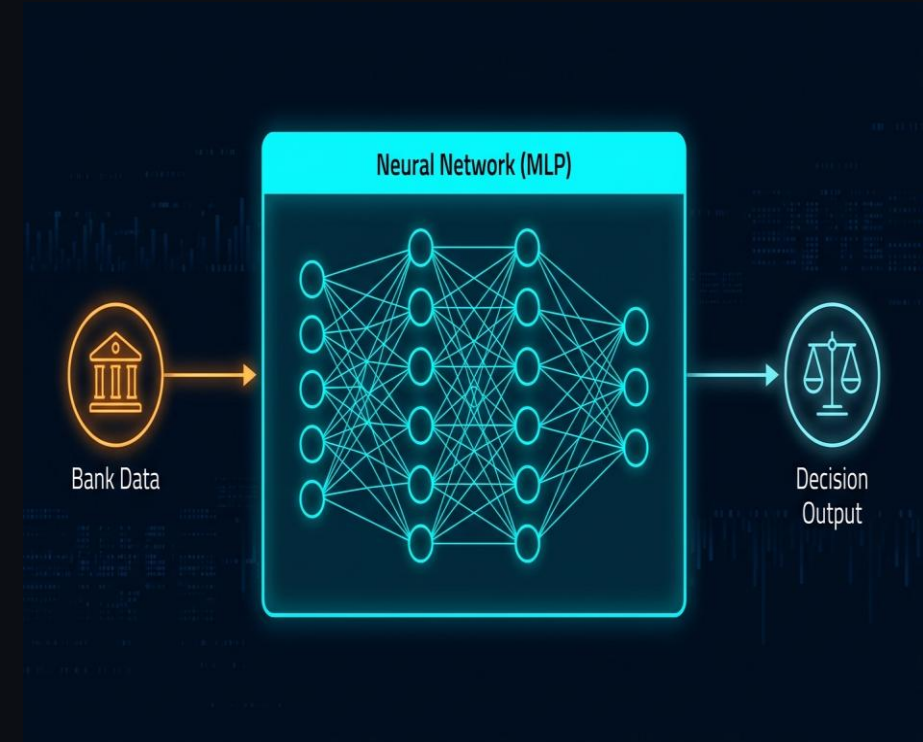


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K. N. Toosi University of Technology

Outline

- ▶ **Section 1: Introduction & Objectives**
- ▶ **Section 2: Data Exploration & Preprocessing**
- ▶ **Section 3: MLP Neural Network Modeling**
- ▶ **Section 4: Conclusion**



1. Dataset Structure & Statistics



Global Dataset Overview:

- Total Records: 10,000 active instances
- Feature Space: 14 Columns
- Missing Values: 0 (Clean Dataset)



Key Variables:

- Target: 'Exited' (0 = Retained, 1 = Churned)
- Primary Inputs: Age, Balance, EstimatedSalary...

Dataset Overview: 10,000 Customer Records

Column Name	Data Type
'RowNumber'	int64
'CustomerId'	int64
'Surname'	object
'CreditScore'	int64
'Geography'	object
'Gender'	object
'Age'	int64
'Tenure'	int64
Balance	float64
NumOfProducts	int64
HasCrCard	int64
IsActiveMember	int64
EstimatedSalary	float64
Exited	int64

Memory Usage: 1.4+ MB

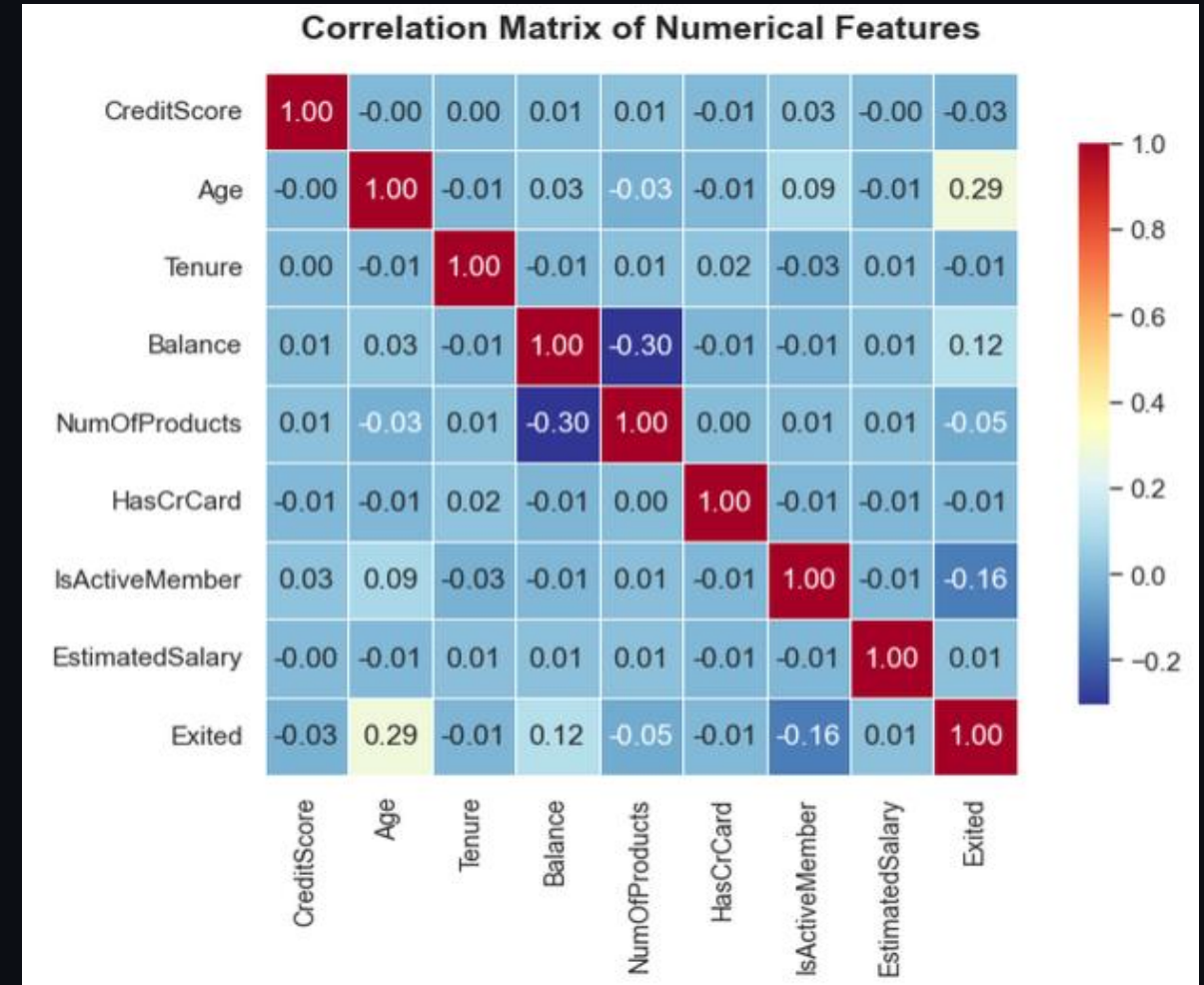
2. EDA (Exploratory Data Analysis)

⚠ Target Class Imbalance:

- Retained (0): ~80% of dataset
- Churned (1): ~20% of dataset

📊 Correlation Insights:

- 'Age' exhibits the highest positive correlation.



3. Data Preprocessing Pipeline

⚡ Transformation Steps:

- └─ Categorical Encoding: One-Hot Encoding
- └─ Feature Scaling: Standardization (Z-score)
 - $z = (x - \mu) / \sigma$
- └─ Dimensionality: Final input vector = 11 features.

📊 Dataset Splitting:

- └─ Training Set: 80%
- └─ Testing Set: 20% (Stratified)

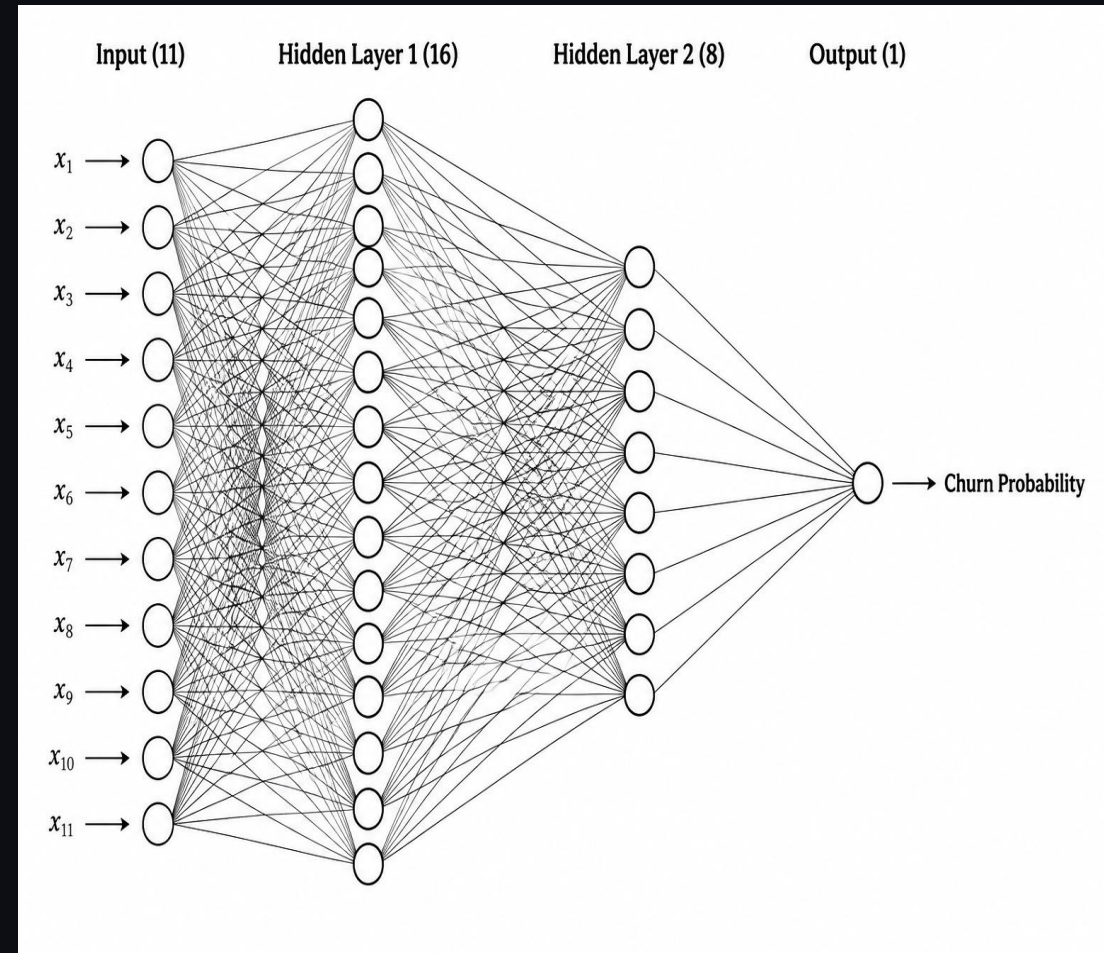
4. MLP Architecture

🧠 Network Topology:

- └─ Input Layer: 11 Neurons
- └─ Hidden Layer 1: 16 Neurons (ReLU)
- └─ Hidden Layer 2: 8 Neurons (ReLU)
- └─ Output Layer: 1 Neuron (Sigmoid)

⚡ Hyperparameters:

- └─ Optimizer: SGD
- └─ Learning Rate: 0.01
- └─ Loss: Binary Cross-Entropy



5. Batch Strategy Analysis

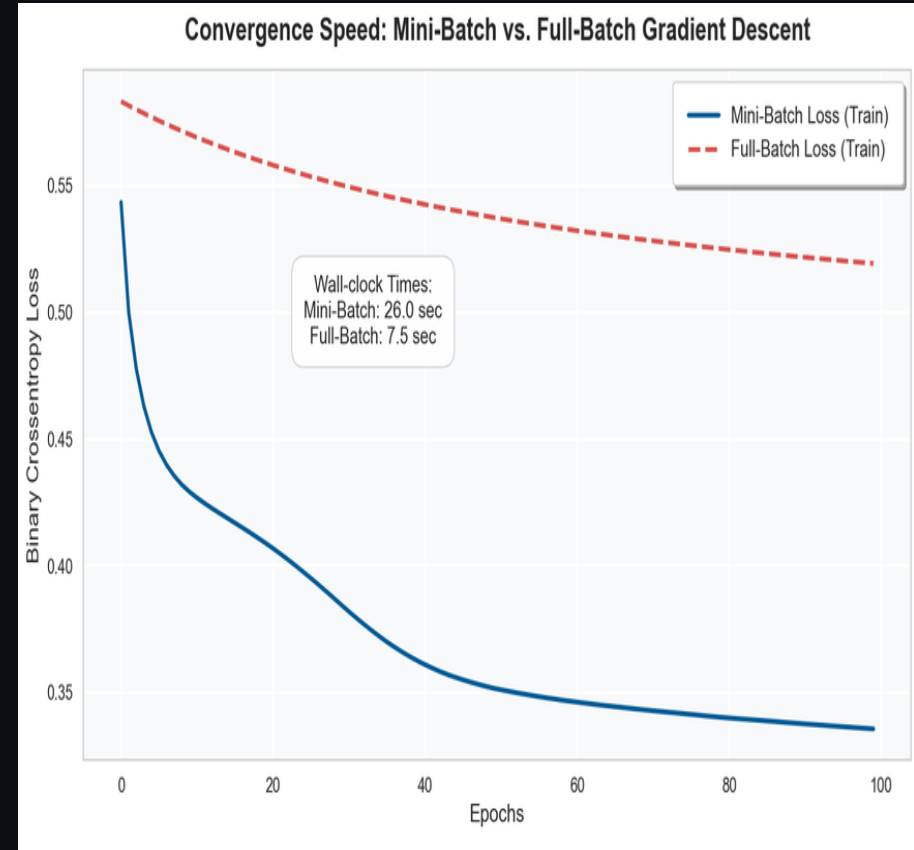
Optimization Benchmarks:

► Mini-Batch Strategy (Size 32):

- └─ Convergence: Faster
- └─ Execution Time: ~26 seconds.

► Full-Batch Strategy:

- └─ Convergence: Slower
- └─ Execution Time: ~7.5 seconds.



6. Model Evaluation & Layer Performance

✓ Performance Metrics (MLP):

└─ ROC-AUC Score: 0.863

📊 Confusion Matrix:

└─ True Negatives (TN): 1543

└─ True Positives (TP): 188

✗ Constraint:

└─ 'Black-box' nature.

